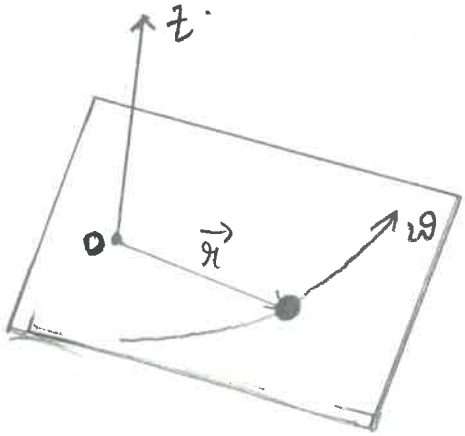
# Angular Momentum & Moment of Inertia




L is the angular momentum of a particle rotating about a fixed pt. as shown in the figure. It is defined as the moment of linear momentum.

$$\vec{L} = \vec{r} \times \vec{p}$$

In Cartesian Coordinates

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{p} = p_x\hat{i} + p_y\hat{j} + p_z\hat{k}$$

$$\therefore \vec{L} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix}$$

$$L_x = yp_z - zp_y$$
$$L_y = zp_x - xp_z$$
$$L_z = xp_y - yp_x$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\frac{d\vec{L}}{dt} = \frac{d(\vec{r} \times \vec{p})}{dt} = \frac{d\vec{r} \times \vec{p}}{dt} + \vec{r} \times \frac{d\vec{p}}{dt}$$

Use $\vec{p} = m\vec{v}$

$$\therefore \frac{d\vec{L}}{dt} = m \cdot \frac{d\vec{r}}{dt} \times \vec{v} + \vec{r} \times m\frac{d\vec{v}}{dt}$$

$$= m \cdot \vec{v} \times \vec{v} + \vec{r} \times \vec{F}$$

($\vec{v} \times \vec{v} = 0$; $m\frac{d\vec{v}}{dt} = \vec{F}$)

$$\therefore \frac{d\vec{L}}{dt} = \vec{r} \times \vec{f} = \vec{\tau}$$ } Torque acting on the system.

If there is no external torque acting on the system, then $\vec{\tau} = 0$

$$\therefore \frac{d\vec{L}}{dt} = 0$$
$$\Rightarrow \vec{L} = \text{Constant}$$ } Conservation of angular momentum.

## Moment of Inertia

$$\vec{L} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v}) \quad \text{using } \vec{p} = m\vec{v}$$

For a body rotating with angular velocity

$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\therefore \vec{L} = m(\vec{r} \times (\vec{\omega} \times \vec{r}))$$

using $A \times (B \times C) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$

$$\vec{L} = m[\vec{\omega}(\vec{r} \cdot \vec{r}) - \vec{r}(\vec{r} \cdot \vec{\omega})]$$

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

& $\vec{\omega} = \omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}$

$$\vec{r} \cdot \vec{r} = x^2 + y^2 + z^2$$

$$\vec{r} \cdot \vec{\omega} = x\omega_x + y\omega_y + z\omega_z$$

$$\therefore \vec{L} = m\Big[(\omega_x\hat{i} + \omega_y\hat{j} + \omega_z\hat{k})(x^2+y^2+z^2) - (x\hat{i}+y\hat{j}+z\hat{k})(x\omega_x + y\omega_y + z\omega_z)\Big]$$

$$\boxed{\vec{L} = m\Big[\{\omega_x(y^2+z^2) - xy\omega_y - xz\omega_z\}\hat{i} + \{\omega_y(x^2+z^2) - yx\omega_x - yz\omega_z\}\hat{j} + \{\omega_z(x^2+y^2) - zx\omega_x - zy\omega_y\}\hat{k}\Big]}$$

$$\therefore \vec{L}_x = m(y^2+z^2)\underline{\omega_x} - mxy\underline{\omega_y} - mxz\underline{\omega_z}$$

$$\vec{L}_y = -mxy\underline{\omega_x} + m(x^2+z^2)\underline{\omega_y} - myz\underline{\omega_z}$$

$$\vec{L}_z = -mxz\underline{\omega_x} - myz\underline{\omega_y} + m(x^2+y^2)\underline{\omega_z}$$

$$\therefore \vec{L}_x = I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z$$

$$\vec{L}_y = I_{yx}\omega_x + I_{yy}\omega_y + I_{yz}\omega_z$$

$$\vec{L}_z = I_{zx}\omega_x + I_{zy}\omega_y + I_{zz}\omega_z$$

Where $I_{xx}, I_{xy}, I_{xz}, I_{yx}, I_{yy}, I_{yz}, I_{zx}, I_{zy}$, $I_{zz}$ are the moment of Inertia.

In short $\vec{I}$ can be written as

$$\begin{bmatrix} L_x \\ L_y \\ L_z \end{bmatrix} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

$$\therefore \vec{L} = \underline{I}\,\vec{\omega}$$

$$I_{xx} = m(y^2 + z^2) \qquad I_{xy} = I_{yx} = -mxy$$

$$I_{yy} = m(x^2 + z^2) \qquad I_{xz} = I_{zx} = -mxz$$

$$I_{zz} = m(x^2 + y^2) \qquad I_{yz} = I_{zy} = -myz$$

Now if there are n particles, the same can be written as

$$I_{xx} = \sum_{i=1}^{n} m_i (y_i^2 + z_i^2) \qquad I_{xy} = I_{yx} = -\sum_{i=1}^{n} m_i x_i y_i$$

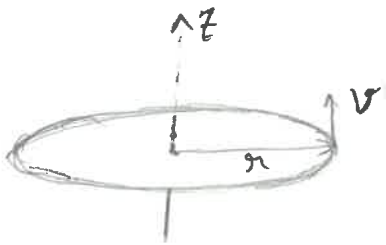
$$I_{yy} = \sum_{i=1}^{n} m_i (x_i^2 + z_i^2) \qquad I_{xz} = I_{zx} = -\sum_{i=1}^{n} m_i x_i z_i$$

$$I_{zz} = \sum_{i=1}^{n} m_i (x_i^2 + y_i^2) \qquad I_{yz} = I_{zy} = -\sum_{i=1}^{n} m_i y_i z_i$$

& $\vec{L}_{net} = \sum_{i=1}^{n} \vec{r}_i \times \vec{p}_i$ & $\vec{L}_{net} = \underline{I}\,\vec{\omega}$

$$\underline{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix}$$

→ If the particles have uniform circular motion




$$\vec{L} = m\left[\vec{r} \times (\vec{\omega} \times \vec{r})\right]$$

$$= m\left[\vec{\omega}(\vec{r}\cdot\vec{r}) - \vec{r}(\vec{r}\cdot\vec{\omega})\right]$$

for uniform circular motion $\vec{r}$ & $\vec{\omega}$ are $\perp^{r}$ to each other $\therefore \vec{r}\cdot\vec{\omega} = 0$

$$\therefore \boxed{\vec{L} = m r^2 \vec{\omega}}$$

$mr^2 = I \quad \therefore \vec{L} = I\vec{\omega}$

Now $\tau = \dfrac{d\vec{L}}{dt} = \dfrac{d(I\omega)}{dt} = I\dfrac{d\omega}{dt} = I\alpha$

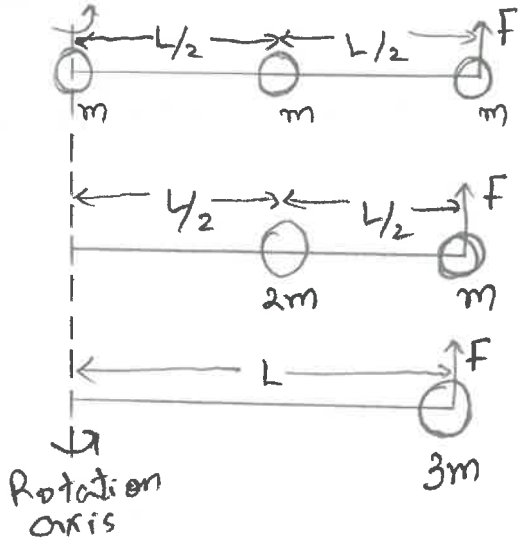
$$\boxed{\tau = I\alpha}$$

↳ 2nd law equivalent in rotational dynamics.

Analogies between rotational & translational motion

| | Translation | Rotation |
|---|---|---|
| displacement | $\vec{s}$ or $\Delta\vec{r}$ | $\theta$ |
| velocity | $\vec{v}$ | $\vec{\omega}$ |
| acceleration | $\vec{a}$ | $\vec{\alpha}$ |
| Cause | $\vec{F}$ | $\vec{\tau}$ |
| Newton's 2nd law | $\Sigma f = m\vec{a}$ | $\Sigma\tau = I\alpha$ |
| Momentum | $\vec{p} = m\vec{v}$ | $\vec{L} = I\vec{\omega}$ |

(9)




3 massless rods A, B & C are free to rotate about an axis as shown in the figure. The same force $\vec{F}$ is applied to the right end of each rod. Objects with different masses are attached to the rods, however each rod has a net 3m masses attached to them. Rank the angular acceleration from largest to smallest.

Sol

$$\tau_{net} = I\alpha$$

$$\alpha = \frac{\tau}{I} = \frac{F \cdot L}{I}$$

for rod A } $I_A = m \cdot 0 + m \cdot \left(\frac{L}{2}\right)^2 + mL^2 = \frac{5}{4} mL^2$

for rod B } $I_B = 2m \cdot \left(\frac{L}{2}\right)^2 + mL^2 = \frac{3mL^2}{2}$

for rod C } $I_C = 3mL^2$

$\because I_A < I_B < I_C$ & F remains the same

$$\alpha_A > \alpha_B > \alpha_C$$

rank of angular acceleration for the three rods.